

Research Notes: Variance, the Law of Large Numbers, and the Central Limit Theorem (Part-1)

From First Principles · A Complete Derivation Log via de Moivre's Original Path

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Methodology: the user drives every step of the derivation; the assistant only executes computations and confirms correctness. Core principle: no presupposed conclusions, no reliance on the known normal-distribution answer, no reverse engineering. The back-and-forth record of mistakes is itself part of the manuscript. e and π must emerge naturally from the derivation, not be reverse-engineered from the target form.

[Entry 001] Opening: the definition of variance and the $n - 1$ controversy

The starting point is the most naive definition:

$$\text{sample variance} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \text{standard deviation} = \sqrt{\text{variance}}$$

Question: why does dividing by $n - 1$ give an estimator whose expectation exactly equals the population variance — the so-called “unbiased estimator” — and how is that proved?

Intuitive explanation (degrees of freedom): when the sample mean $\bar{X}$ stands in for the population mean μ , the data's fluctuation around $\bar{X}$ is naturally slightly smaller than around μ — because $\bar{X}$ is itself computed from the data. The n deviations $(X_i - \bar{X})$ sum identically to zero, so the last one is determined by the first $n - 1$; only $n - 1$ degrees of freedom are independent.

Verdict: this proof needs nothing fancy — elementary algebra suffices.

[Entry 002] The unbiasedness proof, step by step

Let $X_1, \dots, X_n$ be i.i.d. with population mean μ and variance σ^2 ; let $\bar{X}$ be the sample mean. Goal:

$$E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = \sigma^2$$

Rewrite by inserting μ :

$$\sum (X_i - \bar{X})^2 = \sum [(X_i - \mu) - (\bar{X} - \mu)]^2$$

Expand:

$$= \sum (X_i - \mu)^2 - 2(\bar{X} - \mu) \sum (X_i - \mu) + n(\bar{X} - \mu)^2$$

Key identity: $\sum (X_i - \mu) = n(\bar{X} - \mu)$. Substituting:

$$= \sum (X_i - \mu)^2 - 2n(\bar{X} - \mu)^2 + n(\bar{X} - \mu)^2 = \sum (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Take expectations: the first term $= n\sigma^2$; for the second, use the external tool $\text{Var}(\bar{X}) = \sigma^2/n$ (accepted in this entry as a given tool, not proved within this manuscript), so $E[(\bar{X} - \mu)^2] = \sigma^2/n$. Hence

$$E \left[\sum (X_i - \bar{X})^2 \right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$$

Divide both sides by $n-1$. QED.

[Entry 003] Where does the Law of Large Numbers belong: before or after?

Question: should the LLN be proved before the CLT?

Verdict: no — the two are logically independent.

- LLN: $\bar{X}_n \rightarrow \mu$ (a constant; convergence in probability), Chebyshev's inequality suffices.
- CLT: the standardized $\bar{X}_n \rightarrow$ normal distribution (convergence in distribution), requires Stirling expansion (or characteristic functions).

Decision: do the CLT first. The LLN is deferred to Part-2.

[Entry 004] First route choice: the characteristic-function method (rejected)

Standard framework: set $Y_i = (X_i - \mu)/\sigma$, $Z_n = \frac{1}{\sqrt{n}} \sum Y_i$, then

$$\varphi_{Z_n}(t) = [\varphi_Y(t/\sqrt{n})]^n, \quad \varphi_Y(t) = 1 - \frac{t^2}{2} + o(t^2)$$

$$\left[1 - \frac{t^2}{2n} + o(t^2/n)\right]^n \rightarrow e^{-t^2/2}$$

Convergence to the standard normal follows by the continuity theorem.

Reason for rejection: the $1/\sqrt{n}$ scaling and the tool e^{it} are both too clean — they are constructions built backwards from a conclusion already known. $e^{-t^2/2}$ pops out with zero suspense, which amounts to not answering “where does the normal distribution come from” at all. Route abandoned; switch to de Moivre’s original 1733 counting path.

| [Entry 005] Switching to de Moivre’s original path

Toss a fair coin n times (n even), let S_n be the number of heads. Exact probability:

$$P(S_n = k) = \binom{n}{k} \left(\frac{1}{2}\right)^n, \quad k = 0, 1, \dots, n$$

One dead end: the global generating function $(p + q)^n = 1$ — since $p + q = 1$ the whole thing is identically 1, all information is compressed away; useless.

(One pause occurred here — judging that the proof couldn’t be finished within a day, interview prep took priority; resumed later.)

| [Entry 006] Resume: start from the central term of the binomial

The central term:

$$P_0 := P(S_n = n/2) = \binom{n}{n/2} \left(\frac{1}{2}\right)^n = \frac{n!}{[(n/2)!]^2} \cdot 2^{-n}$$

Plug in Stirling’s formula $n! \sim \sqrt{2\pi n} (n/e)^n$:

Denominator: $[\sqrt{\pi n} (n/2e)^{n/2}]^2 = \pi n \cdot (n/2e)^n$ (note: after squaring, the exponent is n , not $2n$ — see Pitfall 1).

$$P_0 \sim \frac{\sqrt{2\pi n} (n/e)^n}{\pi n (n/2e)^n} \cdot 2^{-n} = \frac{\sqrt{2\pi n}}{\pi n} \cdot 2^n \cdot 2^{-n} = \sqrt{\frac{2}{\pi n}}$$

Note: this is where π first emerges naturally — it comes from Stirling’s formula, whose $\sqrt{2\pi}$ in turn comes from the Wallis product / Gaussian-integral normalization constant, not from working backwards from the normal distribution.

| [Entry 007] The neighbor ratio: an “obvious” definition — but useful

The exact (not approximate) neighbor ratio:

$$\frac{P(S_n = k+1)}{P(S_n = k)} = \frac{n-k}{k+1}$$

Let $P_d := P(S_n = n/2 + d)$; then

$$\frac{P_d}{P_{d-1}} = \frac{n/2 - d + 1}{n/2 + d}$$

Dispute and resolution: this ratio is nothing but the definition of binomial coefficients — the coefficients of $(a+b)^n$ are the binomial distribution; no discovery here. But it is still useful: it lets us recur from the central term P_0 to any term,

$$P_d = P_0 \prod_{j=0}^{d-1} \frac{n/2 - j}{n/2 + j + 1}$$

A second doubt, resolved: having already killed the factorials with Stirling for the central term, does writing this product in factorial form re-introduce Stirling? No. The central term uses a Stirling **approximation**; the neighbor ratio is an **exact** algebraic identity. Rewriting the product as factorials is mere notation, not a circular argument.

| [Entry 008] Take logs + Taylor expansion

The product cannot be evaluated directly; take natural logs:

$$\ln P_d = \ln P_0 + \sum_{j=0}^{d-1} \ln \frac{n/2 - j}{n/2 + j + 1}$$

Split:

$$\ln(n/2 - j) = \ln(n/2) + \ln\left(1 - \frac{2j}{n}\right), \quad \ln(n/2 + j + 1) = \ln(n/2) + \ln\left(1 + \frac{2(j+1)}{n}\right)$$

Using $\ln(1+x) = x - x^2/2 + O(x^3)$:

$$\begin{aligned} \ln\left(1 - \frac{2j}{n}\right) &= -\frac{2j}{n} - \frac{2j^2}{n^2} + O\left(\frac{j^3}{n^3}\right) \\ \ln\left(1 + \frac{2(j+1)}{n}\right) &= \frac{2(j+1)}{n} - \frac{2(j+1)^2}{n^2} + O\left(\frac{(j+1)^3}{n^3}\right) \end{aligned}$$

Subtract:

$$= -\frac{2(2j+1)}{n} + \frac{2[(j+1)^2 - j^2]}{n^2} + O\left(\frac{(j+1)^3}{n^3}\right) = -\frac{2(2j+1)}{n} \left(1 - \frac{1}{n}\right) + O\left(\frac{(j+1)^3}{n^3}\right)$$

Leading term $= -\frac{2(2j+1)}{n}$. Note the $1/n^2$ -order term is exactly proportional to the leading term (same $2j+1$ factor) — the structure left over after the quadratic terms cancel.

[Entry 009] Sum over j

Using the identity $\sum_{j=0}^{d-1} (2j+1) = d^2$:

$$\ln P_d = \ln P_0 - \frac{2d^2}{n} + R_{n,d}, \quad R_{n,d} = \frac{2d^2}{n^2} + O\left(\frac{d^4}{n^3}\right)$$

Discipline: do not presuppose the order of magnitude of d here; keep the remainder symbolic and control it once and for all in Entry 016. (At one point the claim “the remainder vanishes when $d = O(\sqrt{n})$ ” was made prematurely and got called out — magnitude judgments before the coordinate map is established are skipped steps; see Pitfall 4.)

[Entry 010] Exponentiate: the bell curve emerges

Dropping the remainder (legitimacy in Entry 016):

$$P_d \approx \sqrt{\frac{2}{\pi n}} e^{-2d^2/n}$$

This is already the embryo of the normal pdf. Key observation: as $n \rightarrow \infty$, if the horizontal axis is made continuous, each integer lattice point d corresponds to one infinitesimal step — d labels “the d -th infinitesimal”. The next step is to turn the discrete sum into an integral.

[Entry 011] Riemann sums and the step size Δx

Set $x = d \cdot \Delta x$, where Δx is an as-yet-undetermined infinitesimal step depending on n .

Why the density must be divided by Δx : $\sum_d P_d$ is a sum over integers d with step 1; after changing to the variable x the step becomes Δx . The standard Riemann-sum form is $\sum f(x_i) \Delta x$, so one must write

$$\sum_d P_d = \sum_d \frac{P_d}{\Delta x} \cdot \Delta x \longrightarrow \int f(x) dx, \quad f(x) = \frac{P_d}{\Delta x}$$

Writing $\sum P_d$ directly as $\int f dx$ loses the Δx factor (Pitfall 5).

[Entry 012] Fixing the scale: the forward derivation (revised entry)

Revision note: in the original derivation log, this step was described as “deriving $\Delta x = 2/\sqrt{n}$ from the normalization condition.” That claim is **false** — Entry 013 verifies that normalization holds for *any* scale parameter, so it cannot pin down Δx ; the eventual choice $\sigma_x = 2$ was in fact matched against the form of $N(0, 1)$, violating the manuscript’s own principle (Pitfall 8). The forward path is as follows — and every ingredient was already in hand.

Read the scale directly off the exponent. Rewrite the result of Entry 010 in Gaussian standard form:

$$e^{-2d^2/n} = \exp\left(-\frac{d^2}{2 \cdot (n/4)}\right) = \exp\left(-\frac{d^2}{2\sigma_d^2}\right), \quad \sigma_d^2 = \frac{n}{4}$$

No external information enters here — we merely rewrite the derived exponent as $-d^2/(2\sigma_d^2)$; the value $\sigma_d^2 = n/4$ is **read off** the exponent.

Independent consistency check: S_n is a sum of n independent Bernoulli(1/2) variables, each toss having variance $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, hence

$$\text{Var}(S_n) = \frac{n}{4}$$

The exact variance agrees **exactly** with the scale σ_d^2 that surfaced in the exponent. Not a coincidence — it is an internal correctness check: the Stirling + Taylor expansion automatically reproduces the exact second moment.

Standardization. The natural dimensionless coordinate is $x = d/\sigma_d$, i.e.

$$x = \frac{2d}{\sqrt{n}}, \quad \Delta x = \frac{2}{\sqrt{n}}$$

The coefficient matches automatically (no tuning). Substituting into the density of Entry 011:

$$f(x) = \frac{P_d}{\Delta x} = \frac{\sqrt{n}}{2} \cdot \sqrt{\frac{2}{\pi n}} e^{-x^2/2} = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Coefficient check: $\frac{\sqrt{n}}{2} \sqrt{\frac{2}{\pi n}} = \frac{1}{2} \sqrt{\frac{2}{\pi}} = \frac{1}{\sqrt{2\pi}}$. Exponent: $-\frac{2d^2}{n} = -\frac{2}{n} \cdot \frac{x^2 n}{4} = -\frac{x^2}{2}$.

The standard-normal coefficient $\frac{1}{\sqrt{2\pi}}$ falls out **automatically** from dividing $P_0 \sim \sqrt{2/\pi n}$ by $\Delta x = 2/\sqrt{n}$ — both obtained by forward derivation, with no parameter ever adjusted to fit a target form.

[Entry 013] Normalization: demoted to a consistency check (revised entry)

The original log used normalization as the device that fixes Δx . The computation shows it cannot do that — but as a **check** it remains valuable.

With a general scale $\sigma_x = \sqrt{n} \Delta x$, the density can be written

$$f(x) = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sigma_x} e^{-2x^2/\sigma_x^2}$$

Using the Gaussian integral $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi/a}$ with $a = 2/\sigma_x^2$:

$$\int_{-\infty}^{\infty} f(x) dx = \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sigma_x} \cdot \sigma_x \sqrt{\frac{\pi}{2}} = 1$$

This holds for every $\sigma_x > 0$. Implications:

1. Normalization is independent of the choice of scale — precisely the statement that “a density stays normalized under change of variables” — so it **cannot** determine Δx .
2. It does, however, validate the Δx -insertion of Entry 011: total probability mass is conserved through the discrete-to-continuous passage; no factor was lost or double-counted.
3. The scale must be fixed by something else — namely Entry 012: read $\sigma_d^2 = n/4$ off the exponent, consistent with $\text{Var}(S_n)$.

[Entry 014] Final conclusion

Writing the standardized variable as $\frac{S_n - n/2}{\sqrt{n/4}}$ (the denominator is the standard deviation of S_n):

$$\lim_{n \rightarrow \infty} P\left(a \leq \frac{S_n - n/2}{\sqrt{n/4}} \leq b\right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Partial sums of the binomial distribution converge to integrals of the normal distribution. The de Moivre version ($p = 1/2$) is proved.

For general p : the center moves to $k = np$ and the standardizing denominator becomes $\sqrt{np(1-p)}$; for general distributions, $\sigma\sqrt{n}$ (Part-2 direction).

[Entry 015] The complete list of pitfalls (8 total)

Pitfall 1: exponent error in the Stirling simplification. Wrong: $[(n/2e)^{n/2}]^2 = (n/2e)^{2n}$. Correct: $= (n/2e)^n$. Consequence: the wrong version spawns a divergent $(2e/n)^n$ term; the correct one cleanly yields $\sqrt{2/\pi n}$. Fix protocol: don't pattern-match against known results — redo the exponents from scratch.

Pitfall 2: mistaking the neighbor ratio for a “discovery”. $P_{k+1}/P_k = (n-k)/(k+1)$ is just the definition of binomial coefficients and carries no information by itself. Its value lies in the recurrence structure: from the central term one can reach any P_d .

Pitfall 3: the characteristic-function method proposed too early. Both e^{it} and $1/\sqrt{n}$ are tools constructed backwards from a known conclusion. Reverse engineering cannot answer “why does the normal distribution look like this.”

Pitfall 4: remainder magnitude judgments introduced too early. Saying “the remainder vanishes for $d = O(\sqrt{n})$ ” before the coordinate map (d versus $\sqrt{n}$) is established is a skipped step. Discipline: keep remainders symbolic; control them all at the end.

Pitfall 5: missing Riemann step size. $\sum P_d / \int f dx$ — one must first insert Δx : $\sum (P_d/\Delta x) \Delta x$, with density $P_d/\Delta x$.

Pitfall 6: presupposing $\Delta x = 1/\sqrt{n}$. This comes from the target form of the standard normal — reverse engineering. See Pitfall 8 for the forward fix.

Pitfall 7: declaring “essentially QED” too early. No conclusion before the quantitative $d \leftrightarrow x$ coordinate map is complete.

Pitfall 8 (found while compiling this manuscript): normalization cannot determine Δx . The original log claimed “ $\Delta x = 2/\sqrt{n}$ follows from normalization,” but normalization equals 1 for every σ_x (densities always stay normalized under change of variables), and the subsequent choice $\sigma_x = 2$ was in substance matched against $N(0, 1)$ — stepping on the manuscript's own reverse-engineering red line. Forward fix: read $\sigma_d^2 = n/4$ directly off the exponent $e^{-2d^2/n} = e^{-d^2/(2 \cdot n/4)}$, consistent with the exact variance $\text{Var}(S_n) = n/4$ (a consistency check, not a presupposition); standardizing $x = d/\sigma_d$ then yields $\Delta x = 2/\sqrt{n}$, with the coefficient $\frac{1}{\sqrt{2\pi}}$ matching automatically. Normalization is demoted to a conservation check.

[Entry 016] Controlling the remainder

The higher-order terms discarded in Entries 008–009 are now controlled in one place. In the central region $d = O(\sqrt{n})$:

- Second-order correction: $\sum_{j=0}^{d-1} \frac{2(2j+1)}{n^2} = \frac{2d^2}{n^2} = O(1/n) \rightarrow 0$
- Third-order remainder: $\sum_{j=0}^{d-1} O\left(\frac{(j+1)^3}{n^3}\right) = O\left(\frac{d^4}{n^3}\right) = O(1/n) \rightarrow 0$

Hence dropping the remainder is legitimate in the limit. The window $d = O(\sqrt{n})$ is exactly the natural scale $\sigma_d = \sqrt{n}/2$ read off in Entry 012 — placing the magnitude judgment *after* the coordinate map is the correct logical order (contrast Pitfall 4).

Side note (heuristic, not rigorous): the finite-variance requirement of the general CLT intuitively corresponds to “the fluctuation window must not blow up.” For Bernoulli the variance is always finite, so this remark is not part of the present proof — only a preview of intuition for the general case.

[Entry 017] Final comparison of the two routes

Characteristic functions: short proof, broad scope (any finite-variance distribution); the drawback is that e^{it} and $e^{-t^2/2}$ look like tools constructed after the answer was known — it cannot answer “where the normal comes from.”

De Moivre (this manuscript): pure counting, using only Stirling’s formula and Taylor expansion. e comes from the growth rate of the factorial ($n! \sim \sqrt{2\pi n}(n/e)^n$), π from Stirling’s normalization constant, and $\sigma_d^2 = n/4$ surfaces naturally in the expanded exponent, matching the exact variance. No distributional shape is presupposed. Drawback: covers only the binomial; generalization takes extra work.

[Entry 018] Historical thread (record only)

- de Moivre (1733): binomial approximated by the normal, $p = 1/2$.
- Laplace (1812): generalized to arbitrary p .
- Lyapunov (1901): generalized to arbitrary distributions, via characteristic functions.

This manuscript follows de Moivre’s original path; no characteristic functions used.

[Entry 019] Directions for Part-2 (notes)

1. General p : neighbor ratio $\frac{n-k}{k+1} \cdot \frac{p}{1-p}$, center $k = np$, variance $np(1-p)$.
2. Full proof of the Law of Large Numbers: the Chebyshev route, independent of the CLT.
3. A rigorous characteristic-function proof of the CLT (continuity theorem) as a companion piece.
4. An independent proof of $\text{Var}(\bar{X}) = \sigma^2/n$ (the tool accepted in Entry 002).

[Entry 020] Final confirmation

Throughout, every directional choice was made by the user; the assistant only executed computations and confirmed correctness. No conclusion was presupposed; the characteristic-function route was rejected as reverse engineering; e and π emerged naturally from Stirling and Taylor; the scale $\Delta x = 2/\sqrt{n}$ was read forward off the exponent structure and checked against the exact variance (the original normalization argument was identified as Pitfall 8 during compilation and corrected). All 8 pitfalls of the back-and-forth are on record.

Appendix: quick-reference variable table

Quantity	Expression	Origin
$P_0 = P(S_n = n/2)$	$\sim \sqrt{2/\pi n}$	Stirling's formula (Entry 006)
$P_d = P(S_n = n/2 + d)$	$\approx P_0 e^{-2d^2/n}$	ratio recurrence + Taylor (Entries 007–010)
σ_d^2	$n/4$	read off the exponent, = $\text{Var}(S_n)$ (Entry 012)
x	$d/\sigma_d = 2d/\sqrt{n}$	standardization (Entry 012)
Δx	$2/\sqrt{n}$	from $x = d \Delta x$ (Entry 012)
Density coefficient	$\frac{\sqrt{n}}{2} \sqrt{\frac{2}{\pi n}} = \frac{1}{\sqrt{2\pi}}$	automatic match, no tuning (Entry 012)
Standardized variable	$(S_n - n/2)/\sqrt{n/4}$	final conclusion (Entry 014)

End